

Tribhuvan University
Patan Multiple Campus
Bachelor in Information Technology (BIT)
Mid-Term Exam-2083

Time: 2 hours (10:00 AM to 12:00 PM)
Bachelor Level/3rd Year/Sixth Semester/Science
Subject: NET Centric Computing (BIT351)

Total Marks: 40
Pass Marks: 16

SET-A
Group A

Long Answer Questions

Attempt any ONE question

1x10=10

1. Explain controllers, action methods, and Action Result types in ASP.NET Core MVC. Design an API controller for Product(ProductId, Name, Price, Category) and implement GET, POST, PUT, and DELETE operations that return appropriate JSON responses.
2. Critically differentiate between an abstract class, interface, and sealed class in C#. Explain when each should be used. Write a program that uses an abstract class with an abstract method and a sealed class to prevent further inheritance.

Group B

Short Answer Questions

Attempt any SIX questions

6x5=30

3. Explain MSIL and CLR and describe their roles in the execution of a .NET application.
4. Differentiate between model binding and model validation. Explain validation using data annotations.
5. Explain Dependency Injection and the IoC container. Differentiate between transient, scoped, and singleton services.
6. What are Tag Helpers? Explain any three commonly used Tag Helpers.
7. Define Cross-Site Scripting and explain its preventive measures.
8. Differentiate between TempData, ViewData, and Session State.
9. Compare ADO.NET and Entity Framework Core.

Date: 2083 - 05 - 02 Set 1

PATAN MULTIPLE CAMPUS

Bachelor Level/Third Year/Six Semester
Bachelor in Information Technology (RSM 354)
(Research Methodology)

Full Marks: 40
Pass Marks: 16
Time: 2 hours

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Section – A

Attempt any ONE questions.

(1 × 10 = 10)

1. What do you mean by concept of research? What is the advantage of concept of research?
2. Define exploratory research design and discuss the major types of qualitative techniques for exploratory research design.

Section – B

Attempt any SIX questions.

(6 × 5 = 30)

3. What do you mean by inductive and deductive theory?
4. Define and discuss case study method. Describe the different steps of case study approach.
5. What do you mean by questions? Write kinds of questions and discuss it.
6. What is sampling. Describe different methods of probability sampling.
7. Define physical scale. Describe different types of physical scale.
8. What is called research report? Write the format of academic research report.
9. Write short note on the following:
 - (a) Inform consent.
 - (b) Non-probability sampling

BEST OF LUCK!!!

Patan Multiple Campus
Bachelor in Information Technology (BIT)
Mid-Term Exam-2083

Time: 2 hours (10:00 AM to 12:00 PM)
Bachelor Level/3rd Year/Sixth Semester/Science
Subject: Multimedia Computing (BIT356)

Total Marks: 40
Pass Marks: 16

SET A

Long Questions:

Attempt Any one

[10*1=10]

1. Differentiate between lossless and lossy compression techniques. Explain their working principles, advantages, disadvantages, and multimedia applications.
2. Discuss the different components of multimedia in detail. Explain how text, graphics, image, audio, video, animation, and interactivity work together in a multimedia application.

Short Questions:

Attempt any Six

[6*5=30]

3. Explain the different ways of presenting text in multimedia applications.
4. Calculate the file size in bytes for a 30-second stereo audio recording sampled at 44.1 kHz with 16-bit resolution.
5. Explain the major video recording formats and standards. Discuss JPEG and MPEG.
6. Explain UI/UX design and discuss UI design process with detail stages.
7. Explain video signal representation in detail. Compare analog and digital representations.
8. Explain the applications of multimedia in detail.
9. Consider the message:

ABBCCDDDD

The frequencies are:

Symbol	Frequency
A	1
B	2
C	3
D	4

Construct the Huffman tree, Generate the Huffman code for each symbol and calculate total bits before coding, after coding and percentage of space save.

END

Patan Multiple Campus
Pre-Board Examination - 2083

Subject: Management Information System
Sixth Semester

Time: 3 hrs.

F.M.: 40

P.M.: 16

SET-A

Group 'A'

Attempt any Two Question.

[10*1=10]

1. What are key challenges of Management Information System? Explain merits and demerits of globalization.
2. What is Technical definition of organization? Differentiate between organization and Information System.

Group 'B'

Attempt any Six questions.

[6*5=30]

3. Explain supply chain management system with example.
4. Differentiate between data mart and data warehouse.
5. Explain business value chain model.
6. What is organizations? Explain features of organizations.
7. Illustrate business intelligence? Explain its features.
8. Define business process reengineering. Explain
9. Explain knowledge based system with example.

Patan Multiple Campus
Bachelor in Information Technology (BIT)
Mid-Term Exam-2083

Time: 2 hours (10:00 AM to 12:00 PM)
Bachelor Level/3rd Year/Sixth Semester/Science
Subject: Database Administration (BIT 352)

Total Marks: 40
Pass Marks: 16

SET – B
Group-A

Long Answer Questions
Answer any One Question:

[1×10 =10]

Group-A

Long Answer Questions
Answer any One Question:

[1×10 =10]

1. Explain architecture of oracle database with suitable diagram. List any four administrative tools available to DBA with explanation for each. [6+4]
2. What are the roles and responsibilities of database administrator? Explain about tablespace and datafile in oracle. How we can add new tablespace and datafiles in oracle database? Explain with suitable commands. [4+2+4]

Group-B

Short Answer Questions

Answer any Six Questions.

[6×5 =30]

3. What is privilege? Write the commands to create user, grant privilege, revoke privilege and drop user in oracle database. [1 + 4]
4. How resource management is done using profile? List any three resource parameters with their significance. [2+3]
5. How RMAN is used for backup and recovery? Explain with example. [5]
6. What do you mean by database tuning? Explain some tuning activities [2+3]
7. Why database auditing is needed? Explain about AUDIT_ADMIN and AUDIT_VIEWER Roles. [2+3]
8. Explain about oracle RAC and oracle ASM. [2.5+2.5]
9. Write short notes on:
 - a. Oracle Net Naming Methods
 - b. MTTR Advisor